

Лабораторная работа №10

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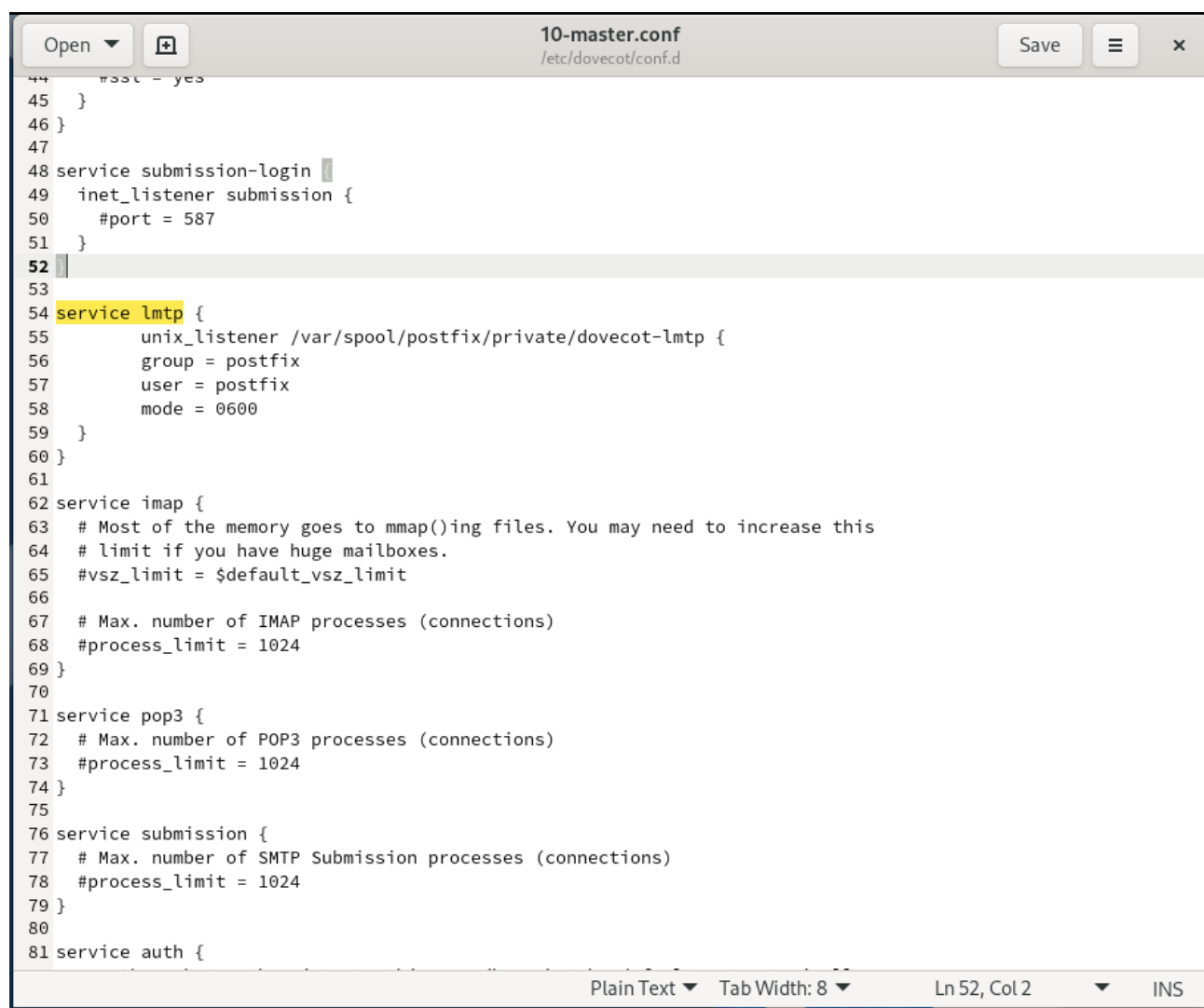
Лабораторная работа №10

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Цель работы

Приобретение практических навыков администрирования сетевых подсистем

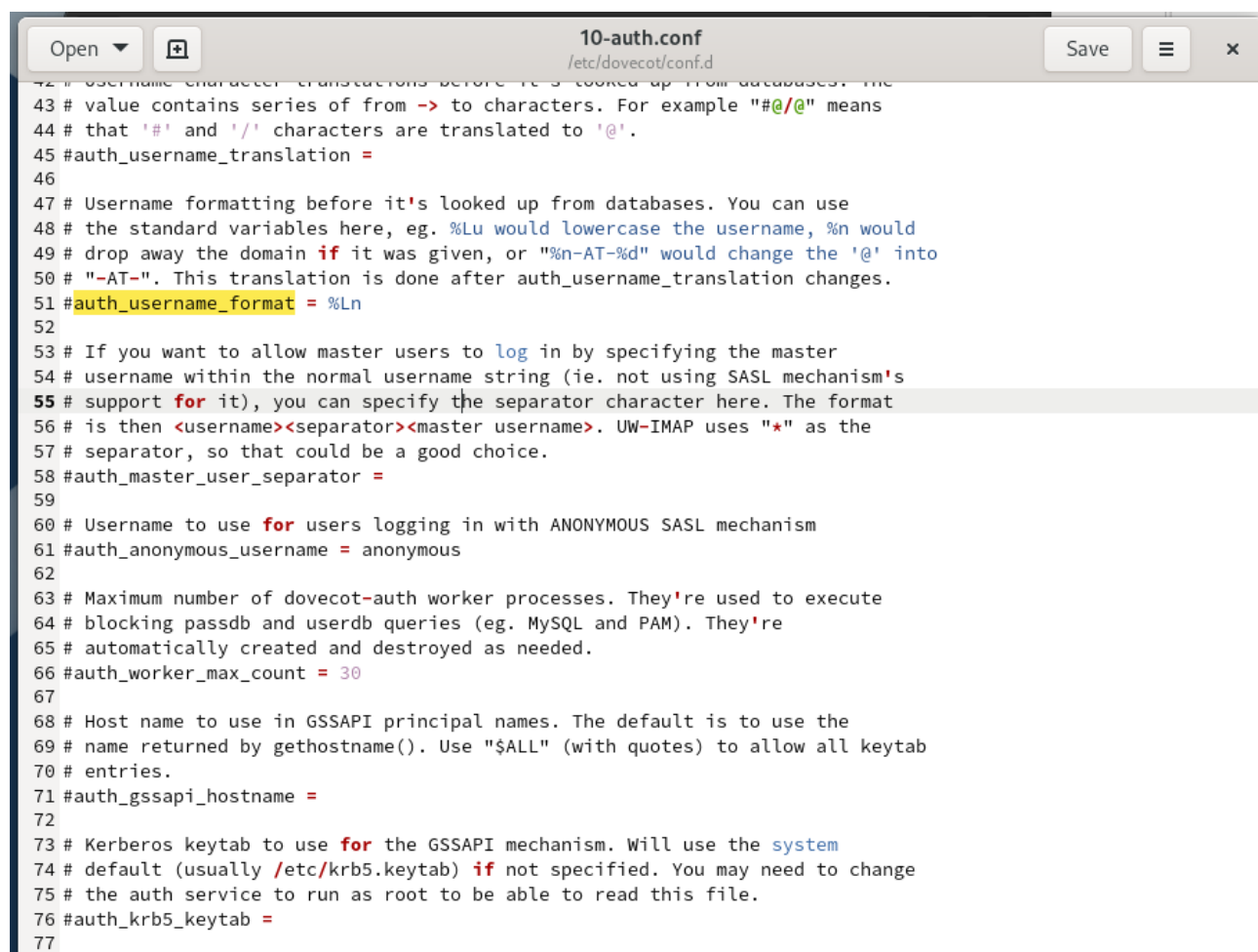
На виртуальной машине server войдём под нашим пользователем и откроем терминал. Перейдём в режим суперпользователя (рис. @fig-1):



```
44 #ssl = yes
45 }
46 }
47
48 service submission-login {
49   inet_listener submission {
50     #port = 587
51   }
52 }
53
54 service lmtp {
55   unix_listener /var/spool/postfix/private/dovecot-lmtp {
56     group = postfix
57     user = postfix
58     mode = 0600
59   }
60 }
61
62 service imap {
63   # Most of the memory goes to mmap()ing files. You may need to increase this
64   # limit if you have huge mailboxes.
65   #vsz_limit = $default_vsz_limit
66
67   # Max. number of IMAP processes (connections)
68   #process_limit = 1024
69 }
70
71 service pop3 {
72   # Max. number of POP3 processes (connections)
73   #process_limit = 1024
74 }
75
76 service submission {
77   # Max. number of SMTP Submission processes (connections)
78   #process_limit = 1024
79 }
80
81 service auth {
```

Рис. 1: Этап 1

В дополнительном терминале запустим мониторинг работы почтовой службы (рис. @fig-2):



```
10-auth.conf
/etc/dovecot/conf.d

42 # Username character translations before it's looked up from databases. The
43 # value contains series of from -> to characters. For example "#@/@" means
44 # that '#' and '/' characters are translated to '@'.
45 #auth_username_translation =
46
47 # Username formatting before it's looked up from databases. You can use
48 # the standard variables here, eg. %Lu would lowercase the username, %n would
49 # drop away the domain if it was given, or "%n-AT-%d" would change the '@' into
50 # "-AT-". This translation is done after auth_username_translation changes.
51 #auth_username_format = %Ln
52
53 # If you want to allow master users to log in by specifying the master
54 # username within the normal username string (ie. not using SASL mechanism's
55 # support for it), you can specify the separator character here. The format
56 # is then <username><separator><master username>. UW-IMAP uses "*" as the
57 # separator, so that could be a good choice.
58 #auth_master_user_separator =
59
60 # Username to use for users logging in with ANONYMOUS SASL mechanism
61 #auth_anonymous_username = anonymous
62
63 # Maximum number of dovecot-auth worker processes. They're used to execute
64 # blocking passdb and userdb queries (eg. MySQL and PAM). They're
65 # automatically created and destroyed as needed.
66 #auth_worker_max_count = 30
67
68 # Host name to use in GSSAPI principal names. The default is to use the
69 # name returned by gethostname(). Use "$ALL" (with quotes) to allow all keytab
70 # entries.
71 #auth_gssapi_hostname =
72
73 # Kerberos keytab to use for the GSSAPI mechanism. Will use the system
74 # default (usually /etc/krb5.keytab) if not specified. You may need to change
75 # the auth service to run as root to be able to read this file.
76 #auth_krb5_keytab =
77
```

Рис. 2: Этап 2

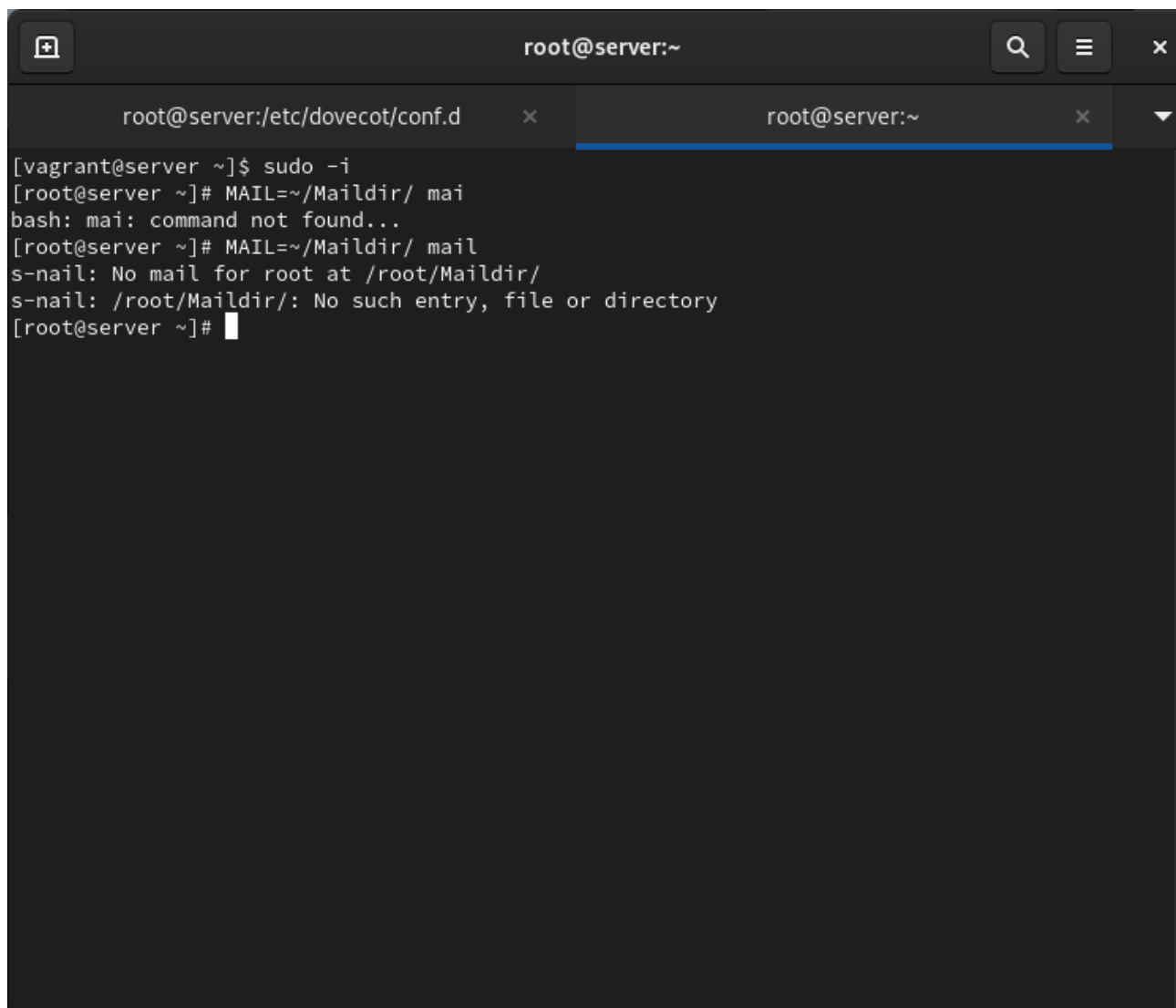
Добавим в список протоколов, с которыми может работать Dovecot, протокол LMTP (рис. @fig-3):

Настроим в Dovecot сервис lmtp для связи с Postfix, определив расположение unix-сокета и права доступа (рис. @fig-4):

Переопределим в Postfix передачу сообщений через заданный unix-сокет (рис. @fig-5):

В файле /etc/dovecot/conf.d/10-auth.conf зададим формат имени пользователя для аутентификации в форме логина без указания домена (рис. @fig-6):

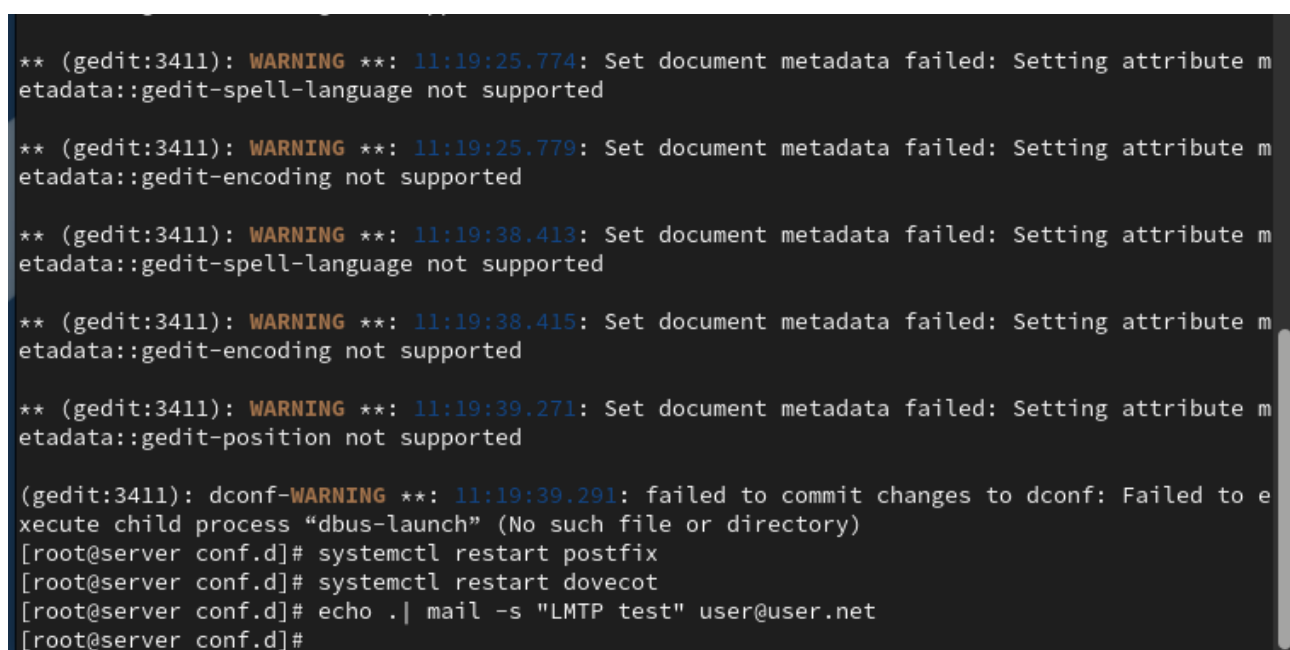
Перезапустим Postfix и Dovecot для применения изменений (рис. @fig-7):



A terminal window titled 'root@server:~' with two tabs: 'root@server:/etc/dovecot/conf.d' and 'root@server:~'. The active tab shows the following commands and output:

```
[vagrant@server ~]$ sudo -i
[root@server ~]# MAIL=~/Maildir/ mai
bash: mai: command not found...
[root@server ~]# MAIL=~/Maildir/ mail
s-nail: No mail for root at /root/Maildir/
s-nail: /root/Maildir/: No such entry, file or directory
[root@server ~]#
```

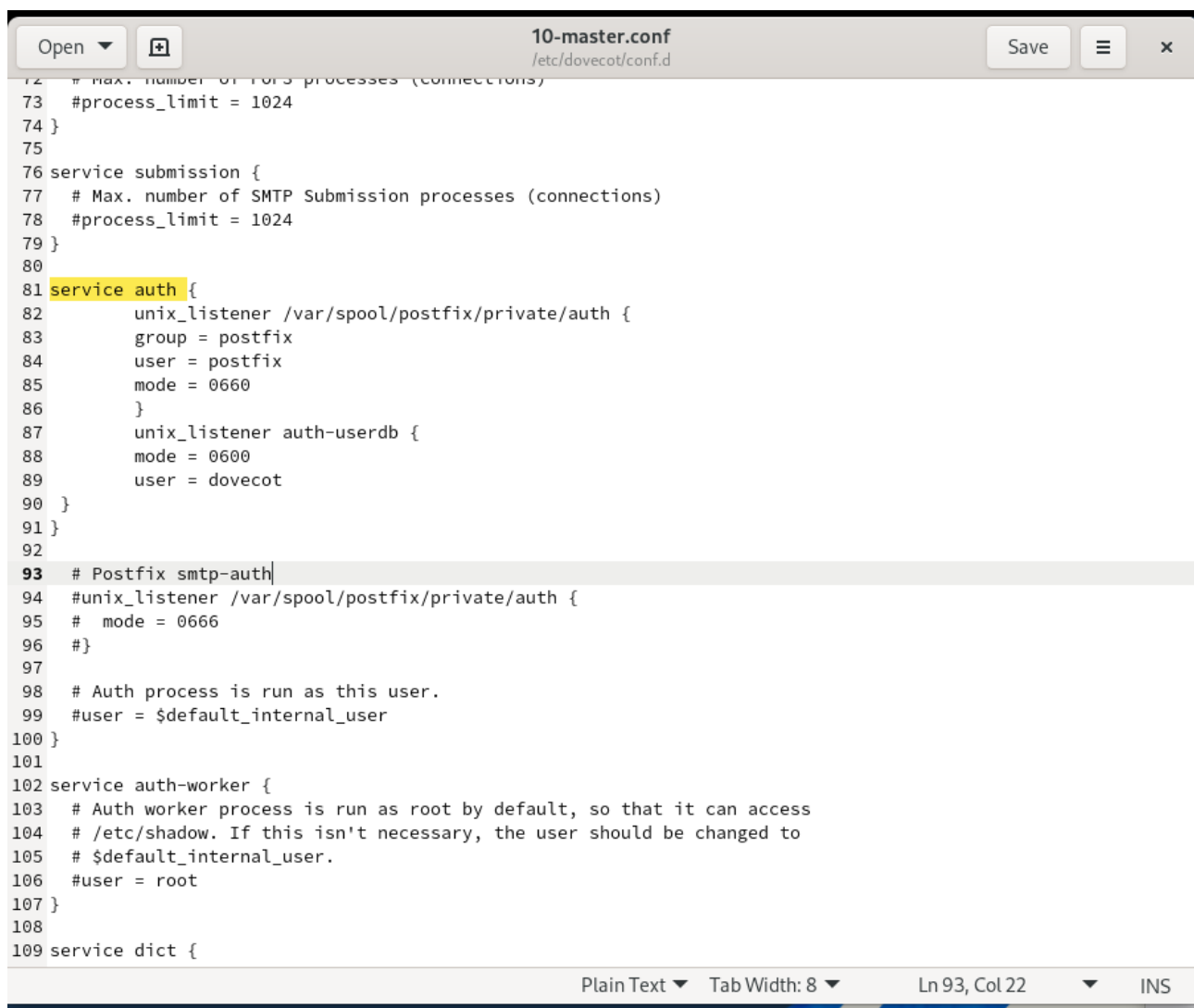
Рис. 3: Этап 3



A terminal window showing the following output:

```
** (gedit:3411): WARNING **: 11:19:25.774: Set document metadata failed: Setting attribute metadata::gedit-spell-language not supported
** (gedit:3411): WARNING **: 11:19:25.779: Set document metadata failed: Setting attribute metadata::gedit-encoding not supported
** (gedit:3411): WARNING **: 11:19:38.413: Set document metadata failed: Setting attribute metadata::gedit-spell-language not supported
** (gedit:3411): WARNING **: 11:19:38.415: Set document metadata failed: Setting attribute metadata::gedit-encoding not supported
** (gedit:3411): WARNING **: 11:19:39.271: Set document metadata failed: Setting attribute metadata::gedit-position not supported
(gedit:3411): dconf-WARNING **: 11:19:39.291: failed to commit changes to dconf: Failed to execute child process "dbus-launch" (No such file or directory)
[root@server conf.d]# systemctl restart postfix
[root@server conf.d]# systemctl restart dovecot
[root@server conf.d]# echo . | mail -s "LMTP test" user@user.net
[root@server conf.d]#
```

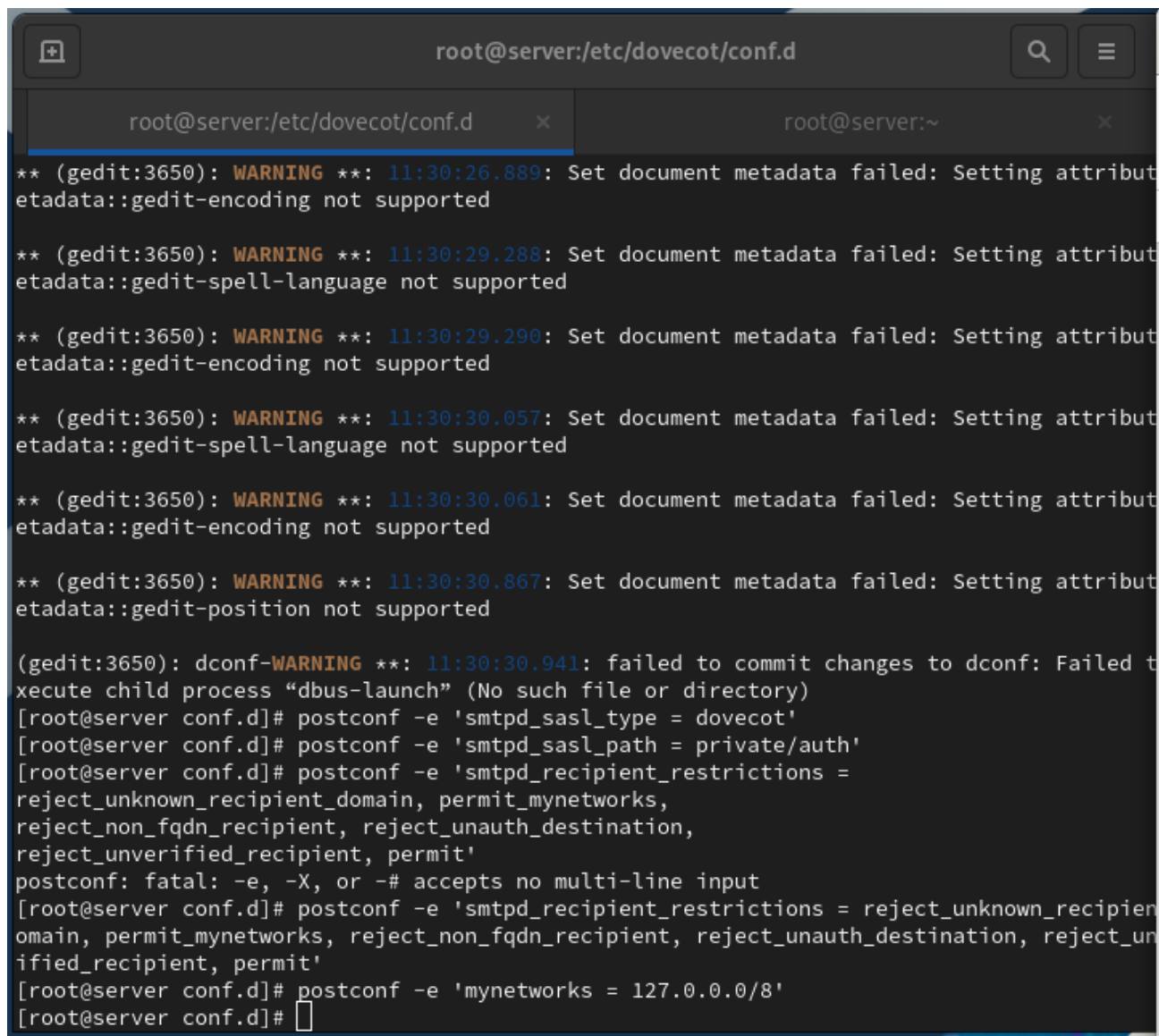
Рис. 4: Этап 4



```
12 # Max. number of POP3 processes (connections)
13 #process_limit = 1024
14 }
15
16 service submission {
17 # Max. number of SMTP Submission processes (connections)
18 #process_limit = 1024
19 }
20
21 service auth {
22     unix_listener /var/spool/postfix/private/auth {
23         group = postfix
24         user = postfix
25         mode = 0660
26     }
27     unix_listener auth-userdb {
28         mode = 0600
29         user = dovecot
30     }
31 }
32
33 # Postfix smtp-auth
34 #unix_listener /var/spool/postfix/private/auth {
35 # mode = 0666
36 #}
37
38 # Auth process is run as this user.
39 #user = $default_internal_user
40 }
41
42 service auth-worker {
43 # Auth worker process is run as root by default, so that it can access
44 # /etc/shadow. If this isn't necessary, the user should be changed to
45 # $default_internal_user.
46 #user = root
47 }
48
49 service dict {
```

Plain Text ▾ Tab Width: 8 ▾ Ln 93, Col 22 ▾ INS

Рис. 5: Этап 5



The screenshot shows a terminal window with a dark theme. The title bar at the top reads "root@server:/etc/dovecot/conf.d". Below the title bar, there are two tabs: "root@server:/etc/dovecot/conf.d" (active) and "root@server:~". The terminal content shows several warning messages from gedit (PID 3650) about failed metadata setting attempts for attributes like "encoding", "spell-language", and "position". Following these, a "dconf-WARNING" message appears, stating "failed to commit changes to dconf: Failed to execute child process 'dbus-launch' (No such file or directory)". The user then enters several "postconf" commands to configure Dovecot's SMTP settings, including "smtpd_sasl_type", "smtpd_sasl_path", and "smtpd_recipient_restrictions". The terminal output shows the configuration being applied line by line. The final command entered is "postconf -e 'mynetworks = 127.0.0.0/8'", and the prompt returns to the root user at the server.

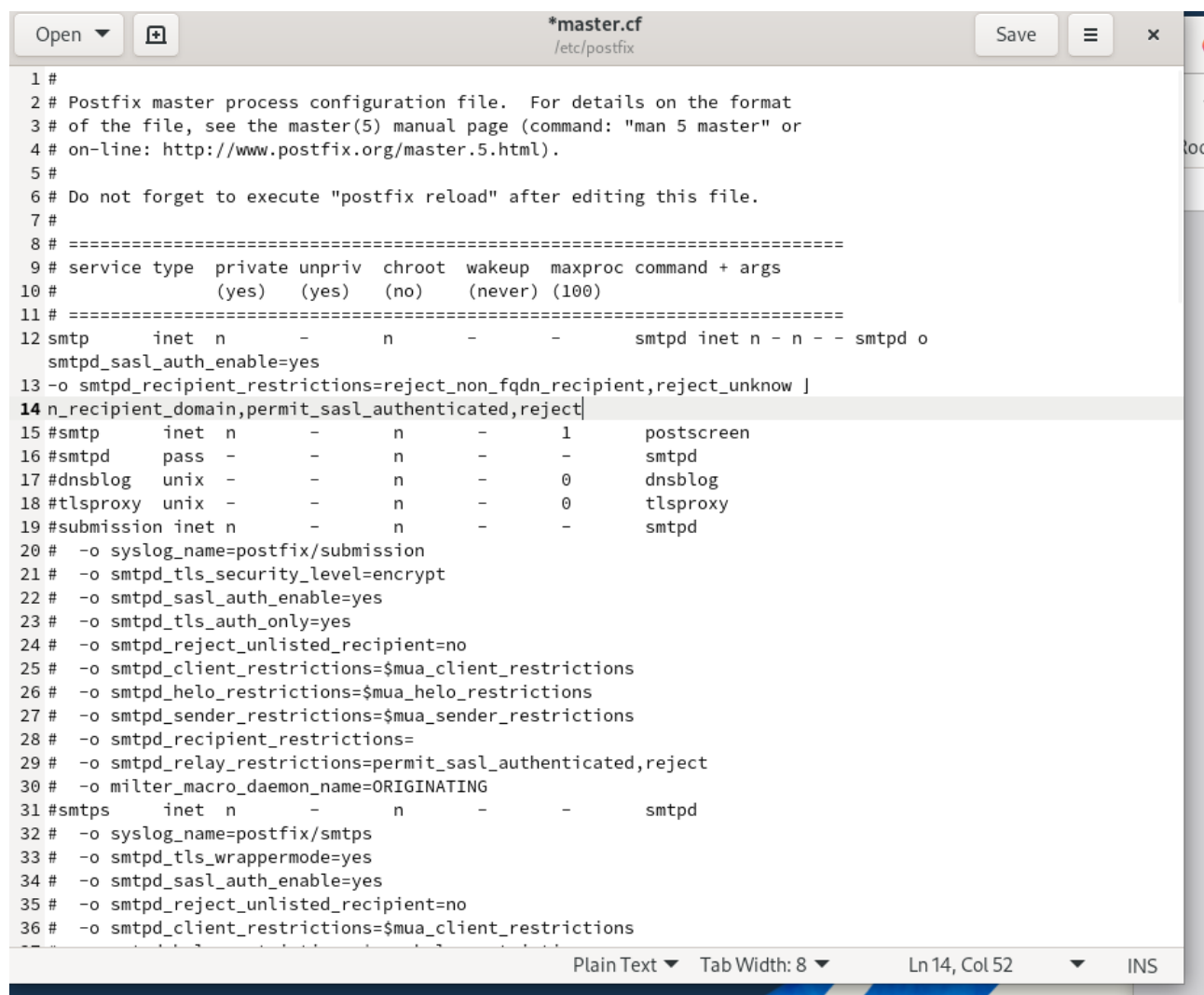
```
root@server:/etc/dovecot/conf.d
** (gedit:3650): WARNING **: 11:30:26.889: Set document metadata failed: Setting attribute etadata::gedit-encoding not supported
** (gedit:3650): WARNING **: 11:30:29.288: Set document metadata failed: Setting attribute etadata::gedit-spell-language not supported
** (gedit:3650): WARNING **: 11:30:29.290: Set document metadata failed: Setting attribute etadata::gedit-encoding not supported
** (gedit:3650): WARNING **: 11:30:30.057: Set document metadata failed: Setting attribute etadata::gedit-spell-language not supported
** (gedit:3650): WARNING **: 11:30:30.061: Set document metadata failed: Setting attribute etadata::gedit-encoding not supported
** (gedit:3650): WARNING **: 11:30:30.867: Set document metadata failed: Setting attribute etadata::gedit-position not supported
(gedit:3650): dconf-WARNING **: 11:30:30.941: failed to commit changes to dconf: Failed to execute child process "dbus-launch" (No such file or directory)
[root@server conf.d]# postconf -e 'smtpd_sasl_type = dovecot'
[root@server conf.d]# postconf -e 'smtpd_sasl_path = private/auth'
[root@server conf.d]# postconf -e 'smtpd_recipient_restrictions = reject_unknown_recipient_domain, permit_mynetworks, reject_non_fqdn_recipient, reject_unauth_destination, reject_unverified_recipient, permit'
postconf: fatal: -e, -X, or -# accepts no multi-line input
[root@server conf.d]# postconf -e 'smtpd_recipient_restrictions = reject_unknown_recipient_domain, permit_mynetworks, reject_non_fqdn_recipient, reject_unauth_destination, reject_unverified_recipient, permit'
[root@server conf.d]# postconf -e 'mynetworks = 127.0.0.0/8'
[root@server conf.d]#
```

Рис. 6: Этап 6

```
1 #
2 # Postfix master process configuration file. For details on the format
3 # of the file, see the master(5) manual page (command: "man 5 master" or
4 # on-line: http://www.postfix.org/master.5.html).
5 #
6 # Do not forget to execute "postfix reload" after editing this file.
7 #
8 # =====
9 # service type  private unpriv  chroot  wakeup  maxproc command + args
10 #              (yes)   (yes)   (no)    (never) (100)
11 # =====
12 smtp          inet  n        -       n       -       -       smtpd
13 #smtp         inet  n        -       n       -       1       postscreen
14 #smtpd        pass  -       -       n       -       -       smtpd
15 #dnsblog      unix  -       -       n       -       0       dnsblog
16 #tlsproxy     unix  -       -       n       -       0       tlsproxy
17 #submission   inet  n        -       n       -       -       smtpd
18 # -o syslog_name=postfix/submission
19 # -o smtpd_tls_security_level=encrypt
20 # -o smtpd_sasl_auth_enable=yes
21 # -o smtpd_tls_auth_only=yes
22 # -o smtpd_reject_unlisted_recipient=no
23 # -o smtpd_client_restrictions=$mua_client_restrictions
24 # -o smtpd_helo_restrictions=$mua_helo_restrictions
25 # -o smtpd_sender_restrictions=$mua_sender_restrictions
26 # -o smtpd_recipient_restrictions=
27 # -o smtpd_relay_restrictions=permit_sasl_authenticated,reject
28 # -o milter_macro_daemon_name=ORIGINATING
29 #smtps        inet  n        -       n       -       -       smtpd
30 # -o syslog_name=postfix/smtps
31 # -o smtpd_tls_wrappermode=yes
32 # -o smtpd_sasl_auth_enable=yes
33 # -o smtpd_reject_unlisted_recipient=no
34 # -o smtpd_client_restrictions=$mua_client_restrictions
35 # -o smtpd_helo_restrictions=$mua_helo_restrictions
36 # -o smtpd_sender_restrictions=$mua_sender_restrictions
37 # -o smtpd_recipient_restrictions=
38 # -o smtpd_relay_restrictions=permit_sasl_authenticated,reject
39 # -o milter_macro_daemon_name=ORIGINATING
```

Рис. 7: Этап 7

Из-под учётной записи своего пользователя отправим письмо с клиента (рис. @fig-8):



```
1 #
2 # Postfix master process configuration file. For details on the format
3 # of the file, see the master(5) manual page (command: "man 5 master" or
4 # on-line: http://www.postfix.org/master.5.html).
5 #
6 # Do not forget to execute "postfix reload" after editing this file.
7 #
8 # =====
9 # service type private unpriv chroot wakeup maxproc command + args
10 # (yes) (yes) (no) (never) (100)
11 # =====
12 smtp inet n - n - - smtpd inet n - n - - smtpd o
13 smtpd_sasl_auth_enable=yes
14 -o smtpd_recipient_restrictions=reject_non_fqdn_recipient,reject_unknow ]
15 n_recipient_domain,permit_sasl_authenticated,reject
16 #smtp inet n - n - 1 postscreen
17 #smtpd pass - - n - - smtpd
18 #dnsblog unix - - n - 0 dnsblog
19 #tlsproxy unix - - n - 0 tlsproxy
20 #submission inet n - n - - smtpd
21 # -o syslog_name=postfix/submission
22 # -o smtpd_tls_security_level=encrypt
23 # -o smtpd_sasl_auth_enable=yes
24 # -o smtpd_tls_auth_only=yes
25 # -o smtpd_reject_unlisted_recipient=no
26 # -o smtpd_client_restrictions=$mua_client_restrictions
27 # -o smtpd_helo_restrictions=$mua_helo_restrictions
28 # -o smtpd_sender_restrictions=$mua_sender_restrictions
29 # -o smtpd_recipient_restrictions=
30 # -o smtpd_relay_restrictions=permit_sasl_authenticated,reject
31 # -o milter_macro_daemon_name=ORIGINATING
32 #smtps inet n - n - - smtpd
33 # -o syslog_name=postfix/smtps
34 # -o smtpd_tls_wrappermode=yes
35 # -o smtpd_sasl_auth_enable=yes
36 # -o smtpd_reject_unlisted_recipient=no
37 # -o smtpd_client_restrictions=$mua_client_restrictions
```

Рис. 8: Этап 8

Посмотрим содержание логов при мониторинге почтовой службы и проверим почтовый ящик пользователя на сервере (рис. @fig-9, @fig-10):

Просмотр логов и почтового ящика

Определим службу аутентификации пользователей в Dovecot и настроим Postfix для использования SASL-аутентификации (рис. @fig-11):

Для проверки работы аутентификации временно запустим SMTP-сервер с возможностью аутентификации (рис. @fig-12):

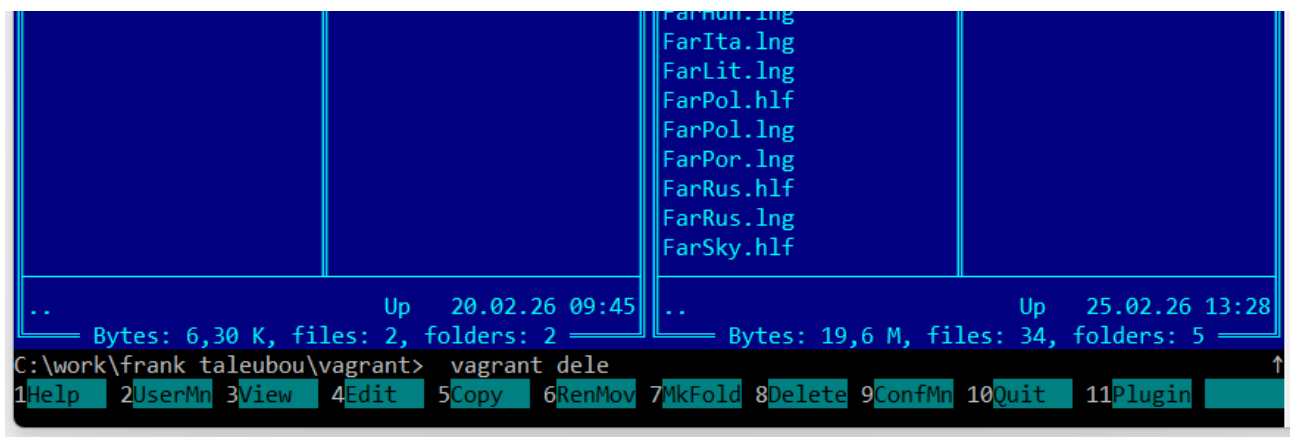


Рис. 9: Этап 9

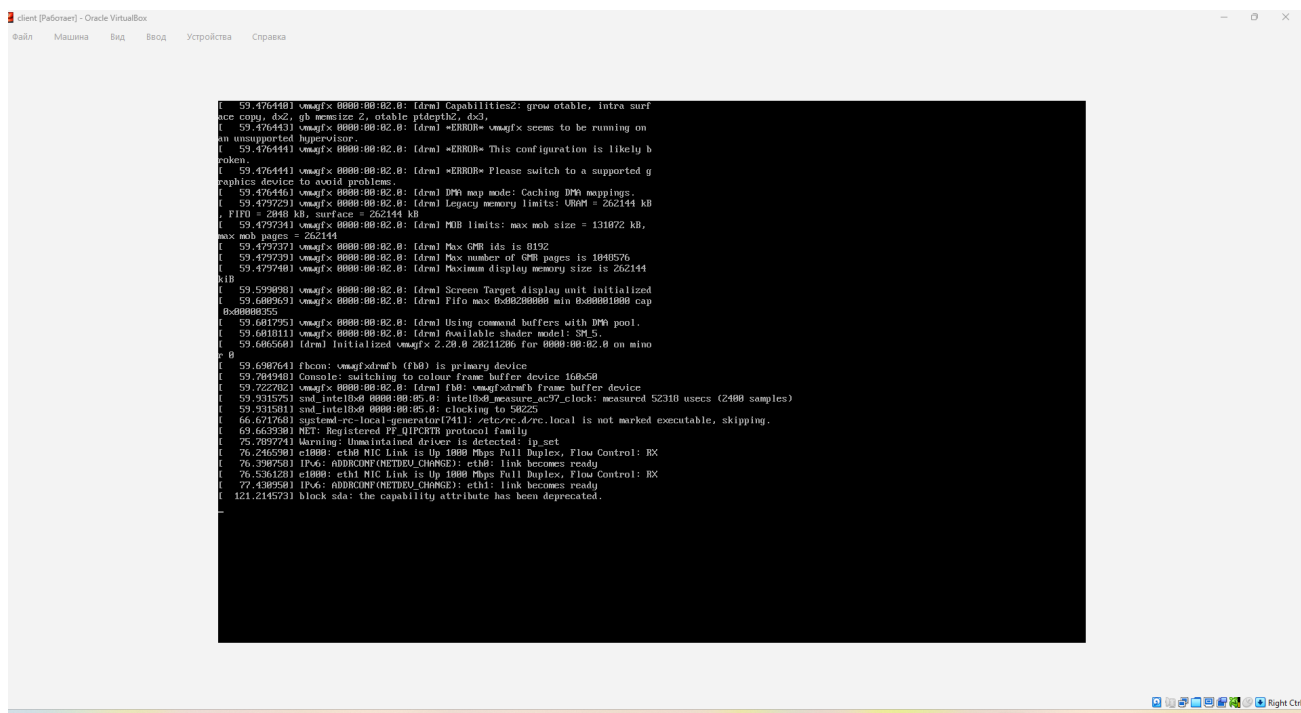


Рис. 10: Этап 10


```
client [Работает] - Oracle VirtualBox
Файл  Машина  Вид  Ввод  Устройства  Справка

Activities  Terminal  Mar 11 12:10  en  [Icons]

root@client:~

[vagrant@client ~]$ sudo -i
[root@client ~]# dnf -y install telnet
Extra Packages for Enterprise Linux 9 - x86_64                27 kB/s | 28 kB      00:01
Extra Packages for Enterprise Linux 9 - x86_64                204 kB/s | 20 MB     01:42
Extra Packages for Enterprise Linux 9 openh264 (From Cisco) - 1.6 kB/s | 993 B     00:00
Rocky Linux 9 - BaseOS                                         5.8 kB/s | 4.3 kB     00:00
Rocky Linux 9 - BaseOS                                         4.9 MB/s | 16 MB     00:03
Rocky Linux 9 - AppStream                                       11 kB/s | 4.8 kB     00:00
Rocky Linux 9 - AppStream                                       4.7 MB/s | 16 MB     00:03
Rocky Linux 9 - Extras                                         561 B/s | 3.1 kB     00:05
Rocky Linux 9 - Extras                                         21 kB/s | 17 kB     00:00
Dependencies resolved.

=====
Package                Architecture      Version           Repository         Size
=====
Installing:
telnet                  x86_64            1:0.17-85.el9     appstream           63 k

Transaction Summary
=====
Install 1 Package

Total download size: 63 k
Installed size: 121 k
Downloading Packages:
telnet-0.17-85.el9.x86_64.rpm                                95 kB/s | 63 kB     00:00
-----
Total                                                         49 kB/s | 63 kB     00:01
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
Preparing...

```

Рис. 11: Этап 11

The screenshot shows a terminal window titled 'client [Работает] - Oracle VirtualBox'. The terminal output is as follows:

```
-----
total                               49 kB/s |  63 kB    00:01
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing      :                                1/1
  Installing     : telnet-1:0.17-85.el9.x86_64    1/1
  Running scriptlet: telnet-1:0.17-85.el9.x86_64    1/1
  Verifying      : telnet-1:0.17-85.el9.x86_64    1/1

Installed:
  telnet-1:0.17-85.el9.x86_64

Complete!
[root@client ~]# print 'vagrant\x00vagrant\x00123456'
bash: print: command not found...
[root@client ~]# telnet server.user.net 25
telnet: server.user.net: Name or service not known
server.user.net: Unknown host
[root@client ~]# telnet server.vagrant.net 25
Trying 13.248.169.48...
telnet: connect to address 13.248.169.48: Connection refused
Trying 76.223.54.146...
telnet: connect to address 76.223.54.146: Connection refused
[root@client ~]# EHLO test
bash: EHLO: command not found...
[root@client ~]# AUTH PLAIN <>
bash: syntax error near unexpected token `newline'
[root@client ~]# AUTH PLAIN dXNlcgB1c2VyADEyMzQ1Ng==
bash: AUTH: command not found...
[root@client ~]#
```

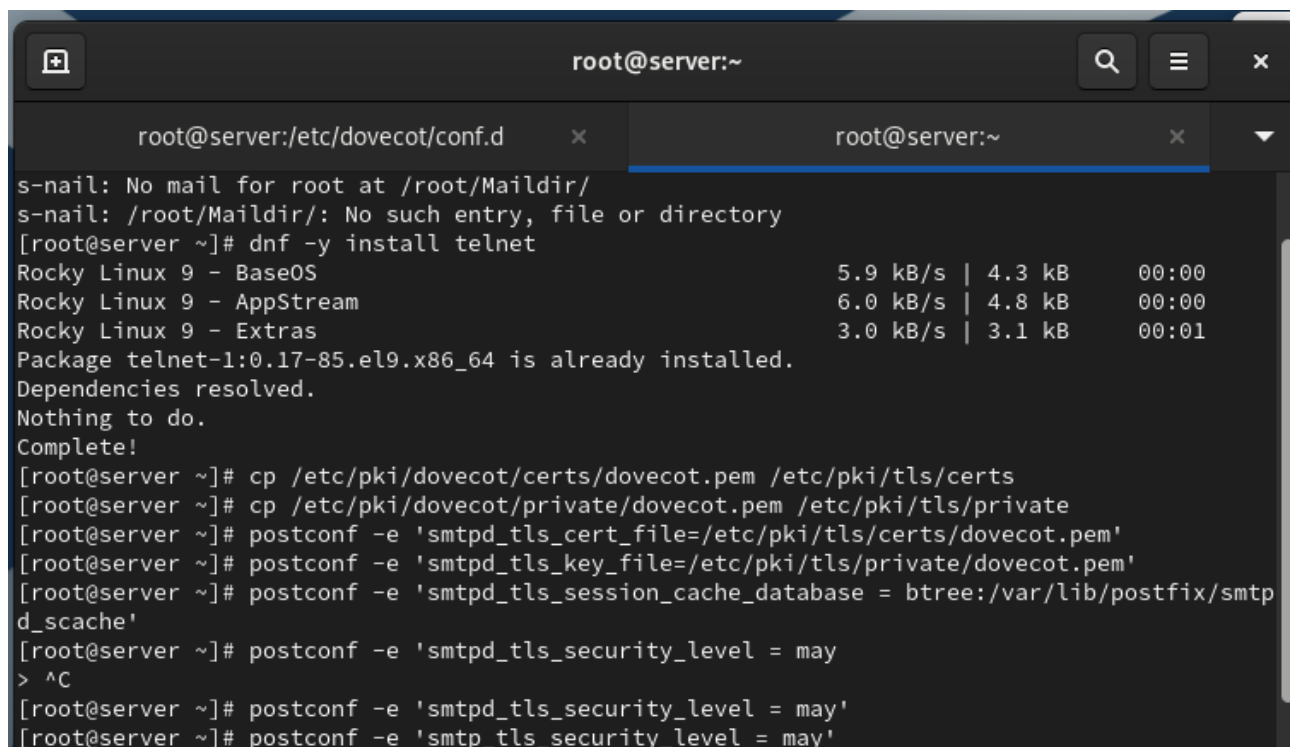
Рис. 12: Этап 12

```
102 #
103 # =====
104 #
105 # Old example of delivery via Cyrus.
106 #
107 #old-cyrus unix - n n - - pipe
108 # flags=R user=cyrus argv=/usr/lib/cyrus-imapd/deliver -e -m ${extension} ${user}
109 #
110 # =====
111 #
112 # See the Postfix UUCP_README file for configuration details.
113 #
114 #uucp unix - n n - - pipe
115 # flags=Fqhu user=uucp argv=uux -r -n -z -a$sender - $nexthop!rmail ($recipient)
116 #
117 # =====
118 #
119 # Other external delivery methods.
120 #
121 #ifmail unix - n n - - pipe
122 # flags=F user=ftn argv=/usr/lib/ifmail/ifmail -r $nexthop ($recipient)
123 #
124 #bsmtp unix - n n - - pipe
125 # flags=Fq. user=bsmtp argv=/usr/local/sbin/bsmtp -f $sender $nexthop $recipient
126 #
127 #scalemail-backend unix - n n - 2 pipe
128 # flags=R user=scalemail argv=/usr/lib/scalemail/bin/scalemail-store
129 # ${nexthop} ${user} ${extension}
130 #
131 #mailman unix - n n - - pipe
132 # flags=FRX user=list argv=/usr/lib/mailman/bin/postfix-to-mailman.py
133 # ${nexthop} ${user}
134 submission inet n - n - - smtpd
135 -o smtpd_tls_security_level=encrypt
136 -o smtpd_sasl_auth_enable=yes
137 -o smtpd_recipient_restrictions=reject_non_fqdn_recipient,reject_unknown
138 n_recipient_domain,permit_sasl_authenticated,reject
```

Рис. 13: Этап 13

На клиенте установим telnet, получим строку для аутентификации и протестируем подключение к SMTP-серверу (рис. @fig-13):

Настроим SMTP-сервер на 587-м порту, разрешим соответствующую службу в межсетевом экране и перезапустим Postfix (рис. @fig-14):



```
root@server:~  
root@server:/etc/dovecot/conf.d x root@server:~ x  
s-nail: No mail for root at /root/Maildir/  
s-nail: /root/Maildir/: No such entry, file or directory  
[root@server ~]# dnf -y install telnet  
Rocky Linux 9 - BaseOS 5.9 kB/s | 4.3 kB 00:00  
Rocky Linux 9 - AppStream 6.0 kB/s | 4.8 kB 00:00  
Rocky Linux 9 - Extras 3.0 kB/s | 3.1 kB 00:01  
Package telnet-1:0.17-85.el9.x86_64 is already installed.  
Dependencies resolved.  
Nothing to do.  
Complete!  
[root@server ~]# cp /etc/pki/dovecot/certs/dovecot.pem /etc/pki/tls/certs  
[root@server ~]# cp /etc/pki/dovecot/private/dovecot.pem /etc/pki/tls/private  
[root@server ~]# postconf -e 'smtpd_tls_cert_file=/etc/pki/tls/certs/dovecot.pem'  
[root@server ~]# postconf -e 'smtpd_tls_key_file=/etc/pki/tls/private/dovecot.pem'  
[root@server ~]# postconf -e 'smtpd_tls_session_cache_database = btree:/var/lib/postfix/smtpd_scache'  
[root@server ~]# postconf -e 'smtpd_tls_security_level = may'  
> ^C  
[root@server ~]# postconf -e 'smtpd_tls_security_level = may'  
[root@server ~]# postconf -e 'smtpd_tls_security_level = may'
```

Рис. 14: Этап 14

Этап выполнения работы №15

Этап выполнения работы №16

Выводы

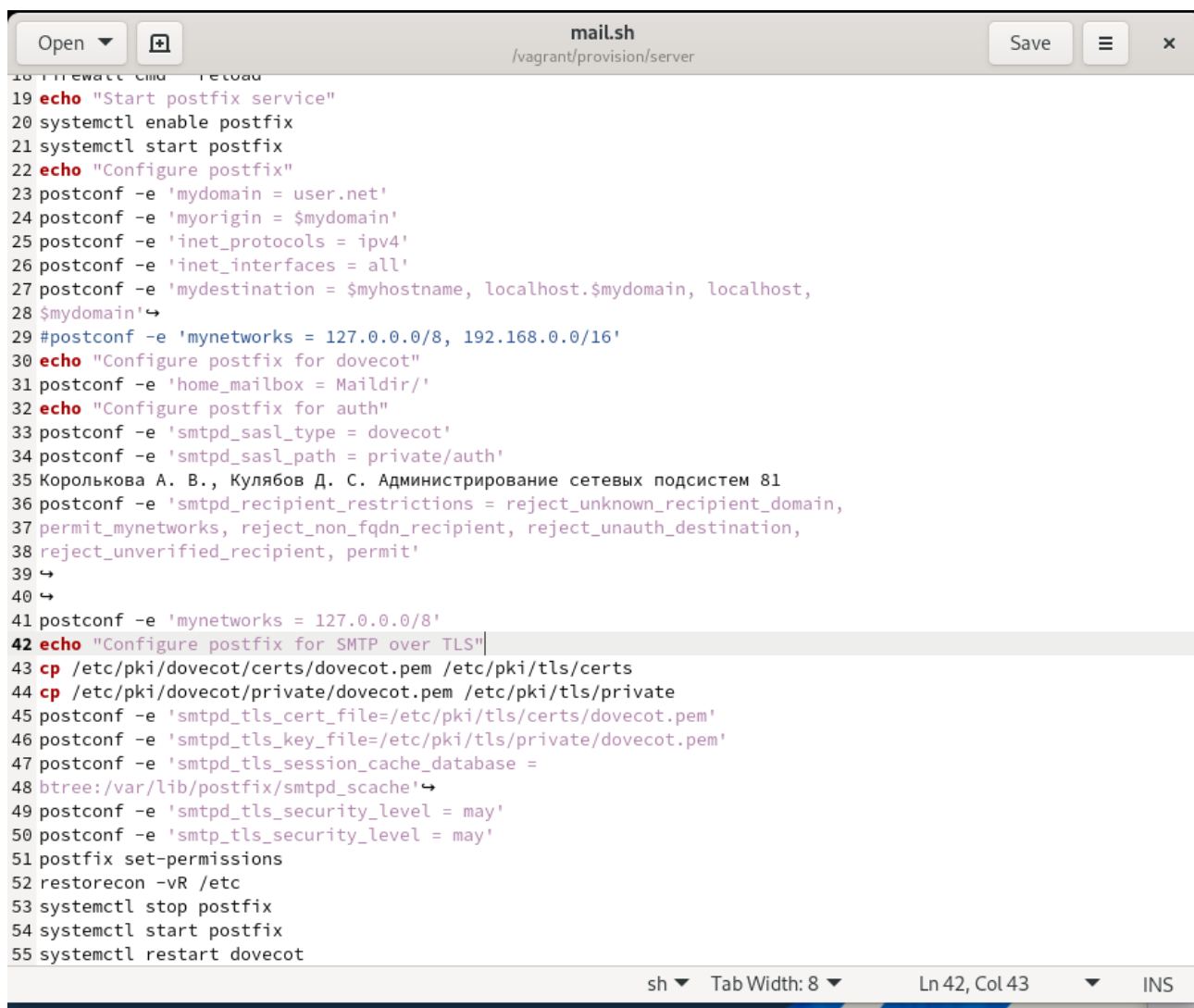
В ходе выполнения лабораторной работы были приобретены практические навыки

Спасибо за внимание!

Вопросы?

```
root@server:~
root@server:/etc/dovecot/conf.d
root@server:~
ix mdns memcache minidlina mongodb mosh mountd mqtt mqtt-tls ms-wbt mssql murmur mysql nbd ne
pula netbios-ns netdata-dashboard nfs nfs3 nmea-0183 nrpe ntp nut openvpn ovirt-imageio ovir
t-storageconsole ovirt-vmconsole plex pncd pmpoxy pmwebapi pmwebapis pop3 pop3s postgresql
privoxy prometheus prometheus-node-exporter proxy-dhcp ps2link ps3netshr ptp pulseaudio pupp
etmaster quassel radius rdp redis redis-sentinel rpc-bind rquotad rsh rsyncd rtsp salt-maste
r samba samba-client samba-dc sane sip sips slp smtp smtp-submission smtps snmp snmptls snmp
tls-trap snmptrap spideroak-lansync spotify-sync squid ssdp ssh ssh-custom steam-streaming s
vdrp svn syncthing syncthing-gui syncthing-relay synergy syslog syslog-tls telnet tentacle t
ftp tile38 tinc tor-socks transmission-client upnp-client vdsrn vnc-server warpinator wbem-ht
tp wbem-https wireguard ws-discovery ws-discovery-client ws-discovery-tcp ws-discovery-udp w
sman wsmans xdmcp xmpp-bosh xmpp-client xmpp-local xmpp-server zabbix-agent zabbix-server ze
rotier
[root@server ~]# firewall-cmd --add-service=smtp-submission
success
[root@server ~]# firewall-cmd --add-service=smtp-submission --permanent
success
[root@server ~]# firewall-cmd --reload
success
[root@server ~]# systemctl restart postfix
Job for postfix.service failed because the control process exited with error code.
See "systemctl status postfix.service" and "journalctl -xeu postfix.service" for details.
[root@server ~]# openssl s_client -starttls smtp -crlf -connect server.vagrant.net:587
801B44331D7F0000:error:8000006F:system library:BI0_connect:Connection refused:crypto/bio/bio
_sock2.c:114:calling connect()
801B44331D7F0000:error:10000067:BI0 routines:BI0_connect:connect error:crypto/bio/bio_sock2.
c:116:
801B44331D7F0000:error:8000006F:system library:BI0_connect:Connection refused:crypto/bio/bio
_sock2.c:114:calling connect()
801B44331D7F0000:error:10000067:BI0 routines:BI0_connect:connect error:crypto/bio/bio_sock2.
c:116:
connect:errno=111
[root@server ~]#
```

Рис. 15: Этап 15



```
18 firewall-cmd --reload
19 echo "Start postfix service"
20 systemctl enable postfix
21 systemctl start postfix
22 echo "Configure postfix"
23 postconf -e 'mydomain = user.net'
24 postconf -e 'myorigin = $mydomain'
25 postconf -e 'inet_protocols = ipv4'
26 postconf -e 'inet_interfaces = all'
27 postconf -e 'mydestination = $myhostname, localhost.$mydomain, localhost,
28 $mydomain'
29 #postconf -e 'mynetworks = 127.0.0.0/8, 192.168.0.0/16'
30 echo "Configure postfix for dovecot"
31 postconf -e 'home_mailbox = Maildir/'
32 echo "Configure postfix for auth"
33 postconf -e 'smtpd_sasl_type = dovecot'
34 postconf -e 'smtpd_sasl_path = private/auth'
35 Королькова А. В., Кулябов Д. С. Администрирование сетевых подсистем 81
36 postconf -e 'smtpd_recipient_restrictions = reject_unknown_recipient_domain,
37 permit_mynetworks, reject_non_fqdn_recipient, reject_unauth_destination,
38 reject_unverified_recipient, permit'
39 ↵
40 ↵
41 postconf -e 'mynetworks = 127.0.0.0/8'
42 echo "Configure postfix for SMTP over TLS"
43 cp /etc/pki/dovecot/certs/dovecot.pem /etc/pki/tls/certs
44 cp /etc/pki/dovecot/private/dovecot.pem /etc/pki/tls/private
45 postconf -e 'smtpd_tls_cert_file=/etc/pki/tls/certs/dovecot.pem'
46 postconf -e 'smtpd_tls_key_file=/etc/pki/tls/private/dovecot.pem'
47 postconf -e 'smtpd_tls_session_cache_database =
48 btree:/var/lib/postfix/smtpd_scache'
49 postconf -e 'smtpd_tls_security_level = may'
50 postconf -e 'smtp_tls_security_level = may'
51 postfix set-permissions
52 restorecon -vR /etc
53 systemctl stop postfix
54 systemctl start postfix
55 systemctl restart dovecot
```

sh Tab Width: 8 Ln 42, Col 43 INS

Рис. 16: Этап 16